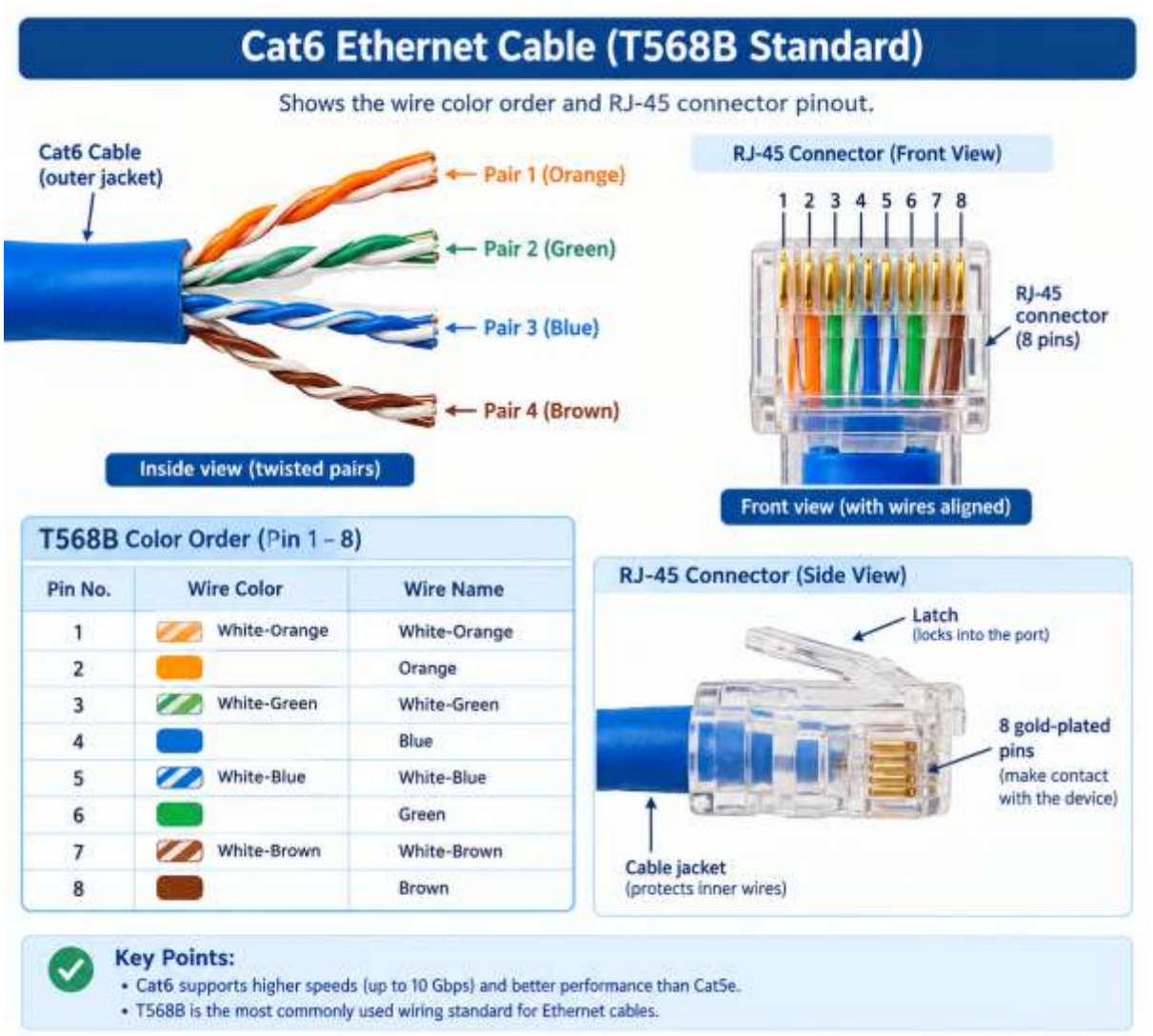
1.

Draw a labeled diagram of a Cat6 Ethernet cable, showing the color order of wires in the T568B standard and labeling the RJ-45 connector pins.

ANS:-



2.

List three types of network cables (such as Cat5e, Cat6, and fiber optic) and for each, mention one real-world app or device (like a WiFi router, Zomato POS terminal, or IPL streaming server) where it would be used, explaining why.

ANS:-

Three Types of Network Cables

Cable Type	**Real-World Device/App**	**Why It Is Used**
Cat5e Ethernet cable	Wi-Fi router connected to a computer	Supports speeds up to 1 Gbps over distances up to 100 metres, making it suitable for ordinary home and office networks.
Cat6 Ethernet cable	Zomato POS terminal in a restaurant	Provides a reliable wired connection for order processing and payment systems, with support for high-speed data transfer.
Fiber-optic cable	IPL streaming server or data center	Transmits data at very high speeds over long distances with low signal loss and immunity to electromagnetic interference.

3.

Watch a YouTube video showing how to crimp an RJ-45 connector onto a network cable, then write down the step-by-step process in your own words and mention two common mistakes to avoid.

ANS:-

A common Ethernet cable termination method uses an RJ-45-compatible 8P8C plug and a crimping tool.

Video tutorial: https://www.youtube.com/watch?v=iUX3qKjI0z4&utm_source=chatgpt.com The steps below follow the T568B wiring standard.

Tools required: Cat5e/Cat6 cable, RJ-45-compatible plugs, wire stripper, crimping tool, and LAN cable tester.

1. Cut the cable: Measure the required length and cut the cable cleanly.
2. Strip the outer jacket: Remove about 2–3 cm of the outer covering without damaging the inner wires.
3. Separate the wires: Untwist and straighten the eight small wires.
4. Arrange the wires: Follow the T568B color sequence from left to right:

White-Orange, Orange, White-Green, Blue, White-Blue, Green, White-Brown, Brown. 5. Trim the wires: Cut all eight wires evenly so they fit into the connector. 6. Insert the wires: Push them fully into the RJ-45 plug in the correct order. Ensure the cable's outer jacket reaches inside the plug. 7. Crimp the connector: Place the plug in the crimping tool and squeeze firmly to secure the contacts. 8. Test the cable: Use a LAN cable tester to check that all eight conductors are connected correctly.

Two common mistakes to avoid:

- Incorrect wire color sequence, which can cause connection failure.
- Wires not inserted fully into the plug, resulting in poor or intermittent connections.

4.

Given the following scenario: You are setting up a gaming zone for an IPL screening event and need to connect multiple screens and consoles. Choose the appropriate cable and connector types for high-speed, reliable connections, and explain your choices.

ANS:-

Cable and Connector Choices for an IPL Screening Gaming Zone

Connection	Recommended cable/connector	Reason
Gaming consoles to network switch	Cat6 Ethernet cable with RJ-45 plugs	Provides fast, stable, low-latency wired networking.
Network switch to router	Cat6 Ethernet cable with RJ-45 plugs	Supports reliable communication with the internet connection.
Gaming consoles to screens	HDMI cable with HDMI connectors	Carries digital video and audio with low delay.
Main network link over a long distance	Fiber-optic cable with compatible optical connectors	Supports high bandwidth over long distances with low signal loss.

Explanation: Cat6 Ethernet is a good choice for connecting gaming devices because it provides reliable, low-latency networking. HDMI cables connect the consoles to screens for high-quality audio and video. Fiber optics are useful for longer network links, although compatible fiber ports or transceivers are required.

5.

Find and share a photo (from your home, college, or online) of any network cable or connector in use (such as behind a WiFi router or at a café's POS counter), and describe what type it is and what devices it connects. Hint: Look for blue or yellow cables with clear plastic ends (RJ-45) or thin round cables for fiber optics.

ANS:-

Type of cable: Ethernet cable (RJ-45 connector), commonly Cat5e or Cat6.

Devices it connects: A Wi-Fi router and a computer, laptop, network switch, or modem, depending on the connection.

Description: The photograph shows a blue Ethernet cable with a transparent RJ-45 connector plugged into a router's network port. The cable carries data between network devices and provides a stable wired connection. Ethernet is commonly used in homes, colleges, offices, and computer laboratories because it offers reliable communication with low latency.

Conclusion: Ethernet cables are important components of computer networks because they connect devices and enable reliable data transmission.

